Paper 2 · Answers & Solutions

2026 · 45 questions

Part 1 · Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	A	C	A	A	B	B	A	B	B	C	D	A	B	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	C	C	B	B	C	A	D	B	D	D	B	D	A	C
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	D	D	A	C	C	B	A	C	C	A	B	A	D	C

Part 2 · Worked Solutions

Section A

1 D

B-
Indices

1 $4^6 = 2^{12}, 8^4 = 2^{12}$

2 $2^{12} \times 2^{12} = 2^{24} = 16^6$, choose D

2 A

B
Linear Equations

1 $\dfrac{a}{x-y} = 3 \Rightarrow a = 3(x-y)$

2 $3x = a + 3y \Rightarrow x = \dfrac{a+3y}{3}$, choose A

3 C

B
Factorization

1 Difference of squares:
$25 - (3x-2y)^2 = [5-(3x-2y)][5+(3x$

2 $= (5-3x+2y)(5+3x-2y)$, choose C

4 A

B
Approximation

1 The first 4 significant figures of 0.0824706 are $8, 2, 4, 7$

2 Rounding gives 0.08247, choose A

5 A

B
Simultaneous Equations

1 From $4a + b = 7a + 3b$: $3a + 2b = 0$; solve with $4a + b = 10$

2 Solving gives $a = 4,\ b = -6$, choose A

6 B

B+
Remainder Theorem

1 Divisible by
$2x + 1 \Rightarrow f(-\frac{1}{2}) = 0 \Rightarrow h = 5$

2 Remainder $= f(-1) = -4 - 5 + 3 = -6$
, choose B

7 B

B
Inequalities

1 $-5x > 21 - 2x \Rightarrow -3x > 21 \Rightarrow x < -7$

2 $6x - 18 < 0 \Rightarrow x < 3$; intersection $x < -7$, choose B

8 A

B+
Quadratic Equations

1 Equal roots $\Rightarrow$ discriminant $= 0$:
$k^2 - 4(8k+36) = 0$

2 $k^2 - 32k - 144 = 0 \Rightarrow k = -4$ or 36, choose A

9 B

A-
Graphs of Quadratic Functions

1 Vertex $x = -\frac{1}{a} > 0$ (as $-1

2 Opens upward, vertex in 4th quadrant, choose B

10 B

B+
Percentages

1 $P = \dfrac{28800}{1.2} = 24000$

2 $T = \dfrac{24000}{0.8} = 30000$, choose B

11 C

A-
Ratios

1 $6(3y - 4x) = 5(2x + y) \Rightarrow 18y - 24x = 1$

2 $13y = 34x \Rightarrow x : y = 13 : 34$, choose C

12 D

A
Variation

1 $z \propto \dfrac{\sqrt{x}}{y}$; new $z = \dfrac{\sqrt{0.64x}}{1.6y} = \dfrac{0.8}{1.6}z$

2 $= 0.5z$, decreased by 50%, choose D

13 A

A-
Mixtures

1 Let Y cost y: $\dfrac{3(45) + 2y}{5} = 39$

2 $135 + 2y = 195 \Rightarrow y = 30$, choose A

14 B

B
Sequences (Patterns)

1 $n\text{th} = 8 + (n-1) \times 6$

2 7th $= 8 + 36 = 44$, choose B

15 **D** — B, Angles

1. From corresponding/alternate angles, $a + b - c = 180°$
2. Only II holds, choose D

16 **D** — A, Cosine Formula

1. $\triangle ABD$: $20^2 + 15^2 = 25^2 \Rightarrow \angle ABD = 90°$
2. $\triangle DBC$: $BC = \sqrt{39^2 - 15^2} = 36$, choose D

17 **C** — A, Parallelograms

1. $\angle BCE = 62°$; $BE = CE \Rightarrow \angle EBC = 62°$
2. $\angle ABE = 118° - 62° = 56°$, choose C

18 **C** — B+, Volume of Prisms

1. Trapezoidal cross-section, area $= \frac{1}{2}(4+8) \times 5 = 30$
2. Volume $= 30 \times 10 = 300$, choose C

19 **B** — A, Sectors

1. Shaded $= \frac{\theta}{360}\pi(40^2 - 32^2) = 72\pi \Rightarrow \theta = 45°$
2. Sector $OAB = 128\pi$; perimeter $OCD = 80 + 10\pi$; all true, choose B

20 **B** — A, Ratios of Areas

1. Let side $= 1$; P, Q on line AG
2. $[DEQP] = \frac{1}{2}$, $[ABCP] = \frac{5}{6} \Rightarrow 3 : 5$, choose B

21 **C** — A, Trigonometry (Resolution)

1. Project AB, BC onto direction AD
2. $AD = AB \sin\alpha + BC \sin\beta$, choose C

22 **A** — A+, Circles and Rhombuses

1. Rhombus $\angle ADC = 124°$; C is centre of circle BDE
2. Using centre-angle relations, $\angle DFE = \frac{3}{2}(124°) - 90° = 96°$, choose A

23 **D** — B, Symmetry

1. Eight hexagons in a symmetric 2×4 arrangement
2. Horizontal and vertical axes $= 2$, choose D

24 **B** — A-, Polygons

1. $(n-2)180 = 2880 \Rightarrow n = 18$
2. Diagonals $= \frac{18 \times 15}{2} = 135$, choose B

25 **D** — A, Perpendicular Lines

1. Product of slopes $= -1$: $\left(-\frac{6}{k}\right)\left(-\frac{3}{4}\right) = -1$
2. $\frac{18}{4k} = -1 \Rightarrow k = -\frac{9}{2}$, choose D

26 **D** — A, Locus and Intersection

1. $AC = BC \Rightarrow C$ on perp. bisector of AB; solve with $x - 2y = 0$
2. $C = (2, 1)$, x-coordinate $= 2$, choose D

27 **B** — A, Equations of Circles

1. $x^2 + y^2 - 4x + 10y + \frac{65}{3} = 0$; centre $(2, -5)$, $r^2 = \frac{22}{3}$
2. $r \neq 14$; origin outside; centre $(2, -5)$; II & III true, choose B

28 **D** — A-, Probability

1. Choosing 3 coins: $C_3^4 = 4$ ways
2. 3 ways give ≥ 13, $P = \frac{3}{4}$, choose D

29 A

A-
Expected Value

1 $E = 50 \cdot \frac{2}{10} + 20 \cdot \frac{3}{10} + 5 \cdot \frac{5}{10}$

2 $= 10 + 6 + 2.5 = 18.5$, choose A

30 C

A
Statistical Measures

1 Mean $77 \Rightarrow a + b + c = 221$; mode $68 \Rightarrow a = b = 68$

2 $c = 85$; median $= \frac{79+85}{2} = 82$, choose C

Section B

31 A — B, L.C.M.

1. Coefficient $\operatorname{lcm}(8,12,20)=120$
2. Highest powers a^5, b^4: $120a^5b^4$, choose A

32 D — A, Logarithms (Linearisation)

1. Line through $(0,2),(4,0)$, slope $=-\frac{1}{2}=\log_9 b$
2. $b=9^{-1/2}=\frac{1}{3}$, choose D

33 D — A, Number Representation

1. D at position 10, E at position 5
2. $=13\times 16^{10}+14\times 16^5$, choose D

34 A — A, Complex Numbers

1. $v=\bar{w}$: $wv=|w|^2$. Here $|w|=1\Rightarrow wv=1$ (rational, I true)
2. Conjugates $\Rightarrow$ equal real parts, opposite imaginary parts; choose A

35 C — A, Linear Programming

1. Optimum of $7y-5x+3$ at a vertex
2. $P(15,30)$ gives max 141, choose C

36 C — A, Geometric Sequences

1. $a_6/a_3=r^3=8\Rightarrow r=2$; all terms rational
2. $S_{99}=3(2^{99}-1)>3\times 10^{28}$; only III, choose C

37 B — A, Trigonometric Graphs

1. Amplitude $=3\Rightarrow a=3$
2. Period $2\Rightarrow b=180$, choose B

38 A — A, Trigonometric Equations

1. Let $s=\sin\theta$: $6s^2+s-2=0\Rightarrow s=\frac{1}{2}$ or $-\frac{2}{3}$
2. Both in $(-1,1)$, 2 roots each, total 4, choose A

39 C — A+, Trigonometry in 3-D

1. Coordinates: P base centre, F above B, $Q=C+8$
2. $\sin\angle PFQ=\dfrac{4\sqrt{61}}{65}$, choose C

40 C — A+, Circles and Tangents

1. Tangents $\Rightarrow \angle BOD=112^\circ$, arc $BD=112^\circ$
2. By the secant-angle relation, $\angle AQB=34^\circ$, choose C

41 A — A, Lines and Circles

1. Substitute $y=2x-8$: $5x^2-52x+128=0$
2. $x_{mid}=\frac{26}{5}$, $y_{mid}=\frac{12}{5}$, choose A

42 B — A, Probability

1. At least 2 tea $=\dfrac{C_2^4C_2^8+C_3^4C_1^8+C_4^4}{C_4^{12}}$
2. $=\dfrac{67}{165}$, choose B

43 A — A, Combinations

1. At most 2 girls $=C_5^{18}+C_1^{12}C_4^{18}+C_2^{12}C_3^{18}$
2. $=8568+36720+53856=99144$, choose A

44 D — A, Stem-and-Leaf and Statistics

1. UQ $=70$ (I false); Ada's $z\approx 1.88<2$ (II true)
2. SD $\approx 11.58<12$ (III false); only II, choose D

45 C

A- Transformation of Variance

1. Each $\times 5$ then $+7$: variance $\times 5^2$, adding a constant has no effect
2. $36 \times 25 = 900$, choose C